



Master Presenter's Manual

for

Programming in C#

Curriculum Code: OV-1508

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Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Understand how C# fits into the .NET Platform
- Analyze the basic structure of a C# application and be able to document, debug, compile, and run simple application
- Create, name, and assign values to variables
- Use common statements to implement flow control, looping, and exception handling
- Create methods (functions and subroutines) that can return values and take parameters
- Explain the basic concepts and terminology of object-oriented programming
- Create, initialize, and destroy objects in a C# application
- Build new C# classes from existing classes
- Create self-contained classes and frameworks in a C# application
- Define operators, use delegates, and add event specifications
- Explore the new features of C# 10.0
- Explain .NET Core Development
- Build Cross-Platform Mobile Apps Using .NET MAUI
- Explore GUI and Desktop Apps with C#

2. Information on Module Allocation

Module	Online (No. of Hrs)	Skill Code
C#	40	1339

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 C#-TL1	Session 2 C#-TL2	Session 3 C#-TL3	Session 4 C#-TL4
2	Session 5 C#-TL5	Session 6 C#-TL6	Session 7 C#-TL7	Session 8 C#-TL8
3	Session 9 C#-TL9	Session 10 C#-TL10	Session 11 C#-TL11	Session 12 C#-TL12
4	Session 13 C#-TL13	Session 14 C#-TL14	Session 15 C#-TL15	Session 16 C#-TL16
5	Session 17 C#-TL17	Session 18 C#-TL18	Session 19 C#-TL19	Session 20 C#-TL20

C#: Programming in C#

TL: Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	PC# – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 1 – Getting Started with C#</u> ➤ Define and describe the .NET Framework ➤ Explain Components of .NET 7.0 ➤ Explain C# language features ➤ Identify various editions of Visual Studio 2022 ➤ Outline the differences between .NET Framework 4.8.1 and .NET 7.0 ➤ Explain the process of running a program on .NET 7.0 <u>Session 2 – Basic Building Blocks in C#</u> ➤ Define and describe variables and data types in C# ➤ Explain comments and XML documentation ➤ Define and describe constants and literals ➤ Describe constant interpolated strings ➤ List the keywords and escape sequences ➤ Explain input and output ➤ Explain different types of statements ➤ Explain the use of different types of operators	<u>Developing Applications with C#</u> SG - Session 1 & 2 XP - Session 1 & 2 TG - Session 1 & 2
2	PC# – TL2	All the topics as listed below from Session 3 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 3 – Programming Constructs and Arrays</u> ➤ Explain selection constructs ➤ Describe loop constructs	<u>Developing Applications with C#</u> SG - Session 3 XP - Session 3 TG - Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain jump statements in C# ➤ Define and describe arrays ➤ Explain different types of arrays ➤ Describe the Array class	
3	PC# – TL3	The Try It Yourself questions of Session 1 to Session 3 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 1-3
4	PC# – TL4	All the topics as listed below from Session 4 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 4 – Classes and Methods in C#</u> ➤ Explain classes and objects ➤ Define and describe methods ➤ List the access modifiers ➤ Explain method overloading ➤ Define and describe constructors and destructors	<u>Developing Applications with C#</u> SG - Session 4 XP - Session 4 TG - Session 4
5	PC# – TL5	All the topics as listed below from Session 5 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 5 – Inheritance and Polymorphism</u> ➤ Define and describe inheritance ➤ Explain method overriding ➤ Define and describe sealed classes ➤ Explain polymorphism	<u>Developing Applications with C#</u> SG - Session 5 XP - Session 5 TG - Session 5
6	PC# – TL6	All the topics as listed below from Session 6 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 6 – Abstract Classes and Interfaces</u> ➤ Define and describe abstract classes ➤ Explain interfaces	<u>Developing Applications with C#</u> SG - Session 6 XP - Session 6 TG - Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Compare abstract classes and interfaces	
7	PC# – TL7	The Try It Yourself questions of Session 4 to Session 6 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 4-6
8	PC# – TL8	All the topics as listed below from Session 7 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 7 – Properties, Indexers, and Record Types</u> ➤ Define properties in C# ➤ Explain properties, fields, and methods ➤ Explain indexers ➤ Describe init only setters and record types	<u>Developing Applications with C#</u> SG - Session 7 XP - Session 7 TG - Session 7
9	PC# – TL9	All the topics as listed below from Session 8 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 8 – Namespaces and Exception Handling</u> ➤ Define and describe namespaces ➤ Define and describe exceptions ➤ Explain the process of throwing and catching exceptions ➤ Explain nested try and multiple catch blocks	<u>Developing Applications with C#</u> SG - Session 8 XP - Session 8 TG - Session 8
10	PC# – TL10	All the topics as listed below from Session 9 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 9 – Events, Delegates, and Collections</u> ➤ Explain delegates ➤ Explain events ➤ Define and describe collections	<u>Developing Applications with C#</u> SG - Session 9 XP - Session 9 TG - Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
11	PC# – TL11	The Try It Yourself questions of Session 7 to Session 9 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 7-9
12	PC# – TL12	All the topics as listed below from Session 10 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 10 – Generics and Iterators</u> ➤ Define and describe generics ➤ Explain creating and using generics ➤ Explain iterators	<u>Developing Applications with C#</u> SG - Session 10 XP - Session 10 TG - Session 10
13	PC# – TL13	All the topics as listed below from Session 11 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 11 – GUI and Connectivity with SQL Database</u> ➤ Define GUI ➤ Identify tools for creating GUI applications ➤ Explain Forms and Controls ➤ Elaborate on the implementation of GUI in C# ➤ Explain database connectivity ➤ Describe SQL Server Database Connectivity using C# ➤ List SQL commands ➤ Explain insert, update, and delete operations in an SQL database	<u>Developing Applications with C#</u> SG - Session 11 XP - Session 11 TG - Session 11
14	PC# – TL14	All the topics as listed below from Session 12 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 12 – Advanced Concepts in C#</u>	<u>Developing Applications with C#</u> SG - Session 12 XP - Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe anonymous methods and extension methods ➤ Explain anonymous types, partial types, and nullable types ➤ Describe system-defined generic delegates ➤ Define lambda expressions ➤ Explain query expressions ➤ Explain parallel and dynamic programming	TG - Session 12
15	PC# – TL15	The Try It Yourself questions of Session 10 to Session 12 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 10-12
16 & 17	PC# – TL16 & TL17	All the topics as listed below from Session 13 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 13 – Building Cross-Platform Mobile Apps Using .NET MAUI</u> ➤ Describe the .NET MAUI architecture ➤ List the features of MAUI ➤ Explain how to install and run the application	<u>Developing Applications with C#</u> SG - Session 13 XP - Session 13 TG - Session 13
18 & 19	PC# – TL18 & TL19	All the topics as listed below from Session 14 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 14 – .NET Development and the Future</u> ➤ Explain what .NET Core is ➤ Explain the history of .NET Core and .NET ➤ Identify differences between .NET Framework and .NET Core ➤ Explain .NET 7.0 and its key features ➤ Describe how to create an application with .NET 7.0 and VS 2022 ➤ Outline .NET 8.0 and the Future	<u>Developing Applications with C#</u> SG - Session 14 XP - Session 14 TG - Session 14

Session No.	Session Title	Session Details	Deliverables' Mapping
20	PC# – TL20	The Try It Yourself questions of Session 13 and Session 14 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 13 & 14

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Student:

Module	Deliverable
Programming in C#	Kit Code: OV-1508-KIT01-01-A Kit Name: Kit of 1 title for OV-1508 Programming Csharp

Deliverables for Faculty:

Module	Deliverable
Programming in C#	Kit Code: OV-1508-KIT01-01-A Kit Name: Kit of 1 title for OV-1508 Programming Csharp
Course	Classroom Presentations Trainer Guide Master PM

6. Library References

➤ Programming C# 10 by Ian Griffiths
➤ C# 10 and .NET 6 – Modern Cross-Platform Development: Build apps, websites, and services with ASP.NET Core 6, Blazor, and EF Core 6 using Visual Studio 2022 and Visual Studio Code 6th ed. Edition by Mark J. Price
➤ Pro C# 10 with .NET 6: Foundational Principles and Practices in Programming 11st ed. Edition by Andrew Troelsen, Phil Japikse

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for PASS

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for CREDIT and

Overall Weighted Marks $\geq 75\%$ qualifies for DISTINCTION

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination (E1)

Marks: 50

Duration: 60 minutes

Type: Objective

Sr. No.	Name of the Subject	Distribution of Marks
1.	Programming in C#	50

Final Examination (R1)

Marks: 50

Duration: 120 minutes

Type: Practical

Sr. No.	Name of the Subject	Distribution of Marks
1.	Programming in C#	50

Evaluation Details

Practical – 50%

OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the module of this course have been described as follows:

8.1 Software Requirements

Microsoft Windows 10
Microsoft Visual Studio 2022 Professional Edition

8.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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